PHYSIOLOGY · MCQ PRACTICE

Oral Cavity & Salivation — MCQ Practice

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GIT BLOCK

Physiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Questions

1. Which cranial nerve carries taste sensation from the anterior two-thirds of the tongue? EASY

A. CN V (trigeminal)
B. CN VII (facial)
C. CN IX (glossopharyngeal)
D. CN X (vagus)
E. CN XII (hypoglossal)

2. A salivary gland secretion described as thin, watery, and rich in amylase is characteristic of which gland? EASY

A. Sublingual gland
B. Minor labial glands
C. Parotid gland
D. Submandibular gland
E. Von Ebner glands

3. Roughly what percentage of saliva is water? EASY

A. ~50%
B. ~70%
C. ~85%
D. ~99%
E. ~40%

4. Which of the following best describes the hormonal regulation of salivary secretion? EASY

A. Primarily controlled by gastrin
B. Primarily controlled by secretin and CCK
C. Under neural control only, with no hormonal regulation
D. Controlled equally by neural and hormonal input
E. Controlled by somatostatin release from D cells

5. A 45-year-old undergoes parotidectomy for a tumor, but her submandibular and sublingual glands are untouched. Which statement about her resting salivary flow is most accurate? MEDIUM

A. She will be essentially anhydrous (no saliva) at rest
B. Her resting saliva volume will fall by only a modest amount, since the submandibular gland supplies most resting flow
C. Her saliva will become markedly more acidic at all flow rates
D. She will lose all antimicrobial protection in saliva
E. Her saliva amylase content will be unaffected

6. A patient is given a muscarinic antagonist for a bowel procedure and reports a dry mouth afterward. Which receptor and division of the autonomic nervous system does this drug block to cause this effect? MEDIUM

- **A.** Beta-adrenergic receptors, sympathetic division
- **B.** M3 receptors, parasympathetic division
- **C.** Nicotinic receptors, somatic division
- **D.** Alpha-1 receptors, sympathetic division
- **E.** M2 receptors, parasympathetic division on the heart

7. During an anxious moment before an exam, a student notices her mouth feels dry and the little saliva present feels thick. Which autonomic pathway is most likely dominant at this moment? MEDIUM

- **A.** Parasympathetic via CN VII and CN IX
- **B.** Sympathetic via T1–T3 and the superior cervical ganglion
- **C.** Somatic motor via CN XII
- **D.** Parasympathetic via the vagus alone
- **E.** Hormonal, via circulating gastrin

8. Compared with saliva collected during rapid, stimulated flow, saliva collected at a very low, resting flow rate is: MEDIUM

- **A.** More hypotonic, more acidic, and richer in potassium
- **B.** Closer to plasma composition and more alkaline
- **C.** Identical in composition, since ductal modification is flow-independent
- **D.** Higher in sodium and bicarbonate
- **E.** Isotonic and rich in bicarbonate

9. A patient with damage isolated to the otic ganglion would be expected to have reduced parasympathetic secretomotor supply to which gland? MEDIUM

- **A.** Submandibular gland
- **B.** Sublingual gland only
- **C.** Parotid gland
- **D.** Minor labial glands
- **E.** Lacrimal gland

10. A patient loses taste sensation over the posterior third of the tongue after a skull base tumor. Which nerve is most likely affected? MEDIUM

- **A.** CN V
- **B.** CN VII
- **C.** CN IX
- **D.** CN X
- **E.** CN XII

11. Which enzyme in saliva is responsible for beginning fat digestion, and where does it become most active? MEDIUM

- **A.** Alpha-amylase; active in the mouth
- **B.** Lingual lipase; becomes most active in the acidic stomach
- **C.** Pepsin; active in the mouth
- **D.** Kallikrein; active in the duodenum
- **E.** Trypsin; active in the stomach

12. Saliva contains kallikrein, which generates bradykinin locally. What is the physiological purpose of this pathway? MEDIUM

- **A.** Activation of pepsinogen
- **B.** Increasing local blood flow to the gland during secretion
- **C.** Buffering acidic reflux in the esophagus
- **D.** Digesting starch into maltose
- **E.** Providing antimicrobial defense against oral bacteria

13. A researcher measures saliva from a gland and finds it is mucous-rich, low in volume, and thick. Which gland is this most likely from? MEDIUM

A. Parotid
B. Submandibular
C. Sublingual
D. Minor glands only
E. It could not come from any named gland

14. A patient with Sjögren syndrome has autoimmune destruction of salivary and lacrimal gland acinar tissue. Which of the following functions of saliva would be LEAST affected, since it depends mainly on a separate immune mechanism rather than acinar secretion? MEDIUM

A. Lubrication for swallowing
B. Digestion of starch
C. Defense via tonsillar lymphoid tissue rather than salivary IgA
D. Buffering of oral pH
E. Antimicrobial action of lysozyme

15. A patient has a lesion that selectively destroys the ductal epithelium of the submandibular gland while leaving the acinar cells intact. Compared to normal, the final saliva at any given flow rate would most likely be: HARD

A. More hypotonic than normal because Na^+/Cl^- reabsorption still occurs normally
B. Closer to the isotonic, plasma-like composition of the primary acinar secretion, because ductal modification is impaired
C. Unaffected, since acinar cells alone determine final composition
D. More concentrated in bicarbonate than normal
E. Identical to parotid saliva

16. A drug selectively blocks beta-adrenergic receptors without affecting muscarinic receptors. During a stressful event, this patient's saliva would most likely show which change compared to an untreated patient? HARD

A. No parasympathetic-driven watery saliva, since beta-blockade removes all salivary drive
B. Loss of the small-volume, mucous/protein-rich sympathetic component, with parasympathetic watery secretion unaffected
C. Complete cessation of all salivary secretion
D. A shift toward more concentrated, protein-rich saliva
E. Increased amylase secretion due to unopposed parasympathetic tone

17. Two patients each have an isolated lesion: Patient A has CN VII damage proximal to the chorda tympani branch; Patient B has damage limited to CN XII. Which statement correctly distinguishes their deficits? HARD

A. Both lose the mastication reflex equally
B. Patient A loses taste from the posterior tongue; Patient B loses taste from the anterior tongue
C. Patient A loses parasympathetic drive to the submandibular gland and anterior tongue taste; Patient B has a purely motor deficit (tongue movement) with taste intact
D. Patient B loses parasympathetic salivary drive; Patient A has no salivary effect
E. Neither deficit affects salivation, since CN VII and CN XII are both purely motor

18. A patient chews food rapidly while anxious before a meeting, producing high salivary flow but under simultaneous strong sympathetic tone. Compared to high-flow saliva under purely parasympathetic drive, this saliva would most likely be: HARD

A. Identical, since flow rate alone determines composition regardless of which division drives it
B. More hypotonic and acidic than usual for that flow rate
C. Higher volume but relatively more mucous/protein-rich than a purely parasympathetic high-flow state would produce
D. Completely free of mucins
E. Indistinguishable from resting flow

19. A previously healthy patient develops achlorhydria-independent, isolated loss of salivary IgA and lysozyme with preserved fluid volume and electrolyte composition. This pattern is most consistent with selective dysfunction of which component of salivary gland physiology? HARD

A. Ductal ion transport

B. Acinar secretion of antimicrobial proteins, with ductal water/electrolyte handling intact

C. Autonomic secretomotor innervation entirely

D. Kallikrein-bradykinin vasodilation

E. Lingual lipase secretion

20. A student argues that because saliva is always hypotonic, sodium concentration in saliva must always be lower than in plasma regardless of flow rate. Why is this reasoning flawed? HARD

A. Saliva sodium is always higher than plasma sodium at every flow rate

B. Saliva is isotonic, not hypotonic, at high flow rates

C. At high flow, ductal transit time is short so less Na+ is reabsorbed, making saliva sodium approach (though not exceed) plasma levels even though it remains hypotonic overall

D. Sodium concentration in saliva is completely independent of ductal function

E. Potassium, not sodium, determines tonicity at all flow rates

✓ Answer key

Q	Answer	Difficulty	Why
1	**B**	EASY	CN VII (chorda tympani) carries taste from the anterior 2/3 of the tongue.
2	**C**	EASY	The parotid produces purely serous, watery, amylase-rich secretion.
3	**D**	EASY	Saliva is about 99% water.
4	**C**	EASY	Unlike most GI secretions, salivary secretion is under neural control only.
5	**B**	MEDIUM	The submandibular gland supplies ~70% of resting saliva; losing the parotid alone has a modest effect on resting (unstimulated) flow, though it will reduce amylase-rich stimulated flow.
6	**B**	MEDIUM	Parasympathetic ACh acting on M3 muscarinic receptors drives copious watery salivation; blocking M3 causes dry mouth.
7	**B**	MEDIUM	Sympathetic stimulation (T1–T3 → superior cervical ganglion → norepinephrine → beta-adrenergic receptors) produces small-volume, mucous/protein-rich saliva — the "dry, thick" sensation under stress.
8	**A**	MEDIUM	Low flow gives more time for ductal Na+/Cl- reabsorption and K+ secretion, producing hypotonic, acidic, K+-rich saliva; high flow approaches plasma composition.
9	**C**	MEDIUM	CN IX synapses in the otic ganglion, which supplies parasympathetic input to the parotid gland.
10	**C**	MEDIUM	CN IX (glossopharyngeal) carries taste from the posterior 1/3 of the tongue.
11	**B**	MEDIUM	Lingual lipase is secreted in saliva but reaches its greatest activity in the acidic gastric environment.
12	**B**	MEDIUM	Kallikrein cleaves kininogen to bradykinin, a vasodilator that increases blood flow to support active secretion.
13	**C**	MEDIUM	The sublingual gland produces a small volume of thick, mucous-rich secretion.
14	**C**	MEDIUM	Salivary IgA, lysozyme, lactoferrin, lubrication, and starch digestion all depend on acinar secretion and would be impaired; tonsillar lymphoid defense is a separate, non-secretory immune mechanism.
15	**B**	HARD	The acinus secretes an isotonic, plasma-like fluid; ductal cells normally reabsorb Na+/Cl- and secrete K+/HCO3- to make it hypotonic. If ductal function is lost, saliva stays closer to the primary isotonic secretion.
16	**B**	HARD	Sympathetic beta-adrenergic stimulation drives the mucous/protein-rich component; blocking it removes that contribution while parasympathetic M3-driven watery secretion (a separate pathway) continues.
17	**C**	HARD	CN VII (proximal to chorda tympani) carries both anterior 2/3 taste and parasympathetic secretomotor fibers to the submandibular ganglion; CN XII is the purely motor nerve to the tongue and does not carry taste or secretomotor fibers.
18	**C**	HARD	Composition depends on both flow rate (via ductal transit time) and which autonomic division dominates; concurrent sympathetic tone adds a mucous/protein-rich component on top of the flow-rate effect.
19	**B**	HARD	

			Fluid volume and electrolyte composition depend on ductal transport, while antimicrobial proteins like IgA and lysozyme are acinar/glandular products; preserved volume/electrolytes with lost antimicrobial proteins points to selective acinar protein secretion failure rather than global gland or autonomic failure.
20	**C**	HARD	Saliva is hypotonic at every flow rate because the ducts are impermeable to water, but the degree of hypotonicity varies: at high flow, shorter ductal transit time means less Na+/Cl- reabsorption, so Na+ rises toward (but does not reach) plasma levels while the fluid remains net hypotonic.